

Predicting Serious or Fatal Road Collisions in Great Britain

Machine Learning Analysis Report

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Overview

This report addresses a supervised binary classification problem: predicting whether a road collision in Great Britain was serious or fatal, as opposed to slight, from the conditions recorded at the scene. I used the *Road Safety Data: Collisions, 2025* file from the Department for Transport (2026), published as open government data (<https://www.gov.uk/government/statistical-data-sets/road-safety-open-data>). I compared logistic regression, a random forest and gradient boosting, and selected a tuned random forest (Breiman, 2001) built with scikit-learn (Pedregosa et al., 2011).

Dataset Description

The dataset contains one row for every personal injury road collision in 2025 that was reported to the police and recorded in the national STATS19 system, which gives 101,525 rows and 44 columns (Department for Transport, 2026). The columns describe the collision location, road, junction, lighting, weather, road surface, date and time, the number of vehicles and casualties, and the recorded severity. The original severity variable has three values (fatal, serious and slight), and I combined fatal and serious into a single positive class, which is the target variable. This class contains 26,644 collisions, or 26.2 percent of the total. I used 17 scene and time columns as model inputs, and excluded location fields and any field that describes the outcome of the collision.

Modeling Approach

Data preparation. I kept only inputs that describe the scene and would be known when a collision is recorded. I removed the casualty count and the variables derived from severity because including information that is only available because of the outcome is a form of target leakage that makes a model appear better than it would be in use (Kaufman et al., 2012). Unknown values, which the Department codes as -1, 9 or 99, were mapped to one unknown category, and no rows were deleted. Categorical codes were one-hot encoded, speed limit and number of vehicles were standardised, and hour and month were encoded with sine and cosine so that the end of a cycle is next to its start. The data were split into 80 percent for training and 20 percent for testing, keeping the same class proportions in both, and all preprocessing was fitted on the training data only.

Model selection. Logistic regression is an interpretable baseline. A random forest averages many decision trees and can represent interactions between conditions without specifying them in advance, and it makes few assumptions about the shape of relationships (Breiman, 2001). Gradient boosting was included as a further tree ensemble. A dummy classifier that always predicts the majority class provided a floor for comparison. The models were compared with five-fold cross-validation on the training set, and the best model was tuned with a small grid search. Because the positive class is the minority, the models used balanced class weights, a cost-sensitive approach for imbalanced data that avoids altering the test distribution (He & Garcia, 2009).

Evaluation metrics. Accuracy can be misleading when one class dominates, because a model that always predicts the majority class can score highly while detecting nothing of interest (He & Garcia, 2009). I therefore used precision, recall and the F1 score, which describe performance on the positive class (Sokolova & Lapalme, 2009). Recall shows how many serious or fatal collisions are found, which matters most in a safety setting, and precision shows how many alerts are correct. I also used ROC AUC, which measures how well a model ranks positive cases above negative ones, where 0.5 indicates chance performance (Fawcett, 2006), and average precision, because precision-recall measures are more informative than ROC curves when the classes are imbalanced (Saito & Rehmsmeier, 2015). The decision threshold was chosen on the training data only. Variable importance was measured by permutation, which avoids biases that affect importance measures built into tree models (Strobl et al., 2007).

Assumptions. The models assume that collisions are independent of each other, that the training and test collisions come from the same population, and that the recorded severity is an accurate label. Logistic regression additionally assumes that each input has a linear effect on the log odds of a serious or fatal outcome, whereas the random forest does not. These assumptions are discussed further under limitations.

Results

Table 1 shows the cross-validated comparison on the training set. All three models clearly beat the dummy baseline, whose average precision (0.262) equals the share of serious or fatal collisions, but they are close to one another. The random forest had the highest average precision (0.376) and was selected. Tuning changed the cross-validated average precision only from 0.3761 to 0.3766. The threshold that maximised F1 on the training data was 0.44.

Table 1. Five-fold cross-validation results on the training set (notebook Section 4).

Model	F1	Precision	Recall	ROC AUC	Average precision
Dummy baseline	0.000	0.000	0.000	0.500	0.262
Logistic regression	0.423	0.325	0.605	0.617	0.358
Random forest	0.431	0.350	0.559	0.635	0.376
Gradient boosting	0.433	0.344	0.586	0.634	0.374

On the held-out test set (20,305 collisions), the tuned random forest reached a recall of 0.823, a precision of 0.311, an F1 score of 0.451, a ROC AUC of 0.640 and an average precision of 0.371, compared with 0.262 for a model with no skill (Table 2 and Figure 2). These results are very close to the cross-validation results, so there is no sign of overfitting. Accuracy was 0.474, lower than the 0.738 of a model that always predicts slight, because the chosen threshold deliberately flags many slight collisions in order to find serious ones. The confusion matrix shows 4,386 serious or fatal collisions correctly flagged, 943 missed, 9,735 slight collisions wrongly flagged and 5,241 correctly left unflagged.

Table 2. Test-set performance (notebook Section 5).

Metric	Always predict slight	Tuned random forest
Accuracy	0.738	0.474
Precision	0.000	0.311
Recall	0.000	0.823
F1	0.000	0.451
ROC AUC	0.500	0.640
Average precision	0.262	0.371

The most important inputs by permutation importance were the number of vehicles and the trunk road flag, followed by special conditions at the site, hour, urban or rural area and speed limit (Figure 3). Single-vehicle collisions were more often serious or fatal (32.8 percent) than two-vehicle collisions (22.7 percent). Collisions with an unknown trunk road flag were serious or fatal 40.3 percent of the time, against about 25 percent for known values, and most of the unknown cases came from one police force code, which suggests that this input partly reflects how collisions were recorded. Across groups, ROC AUC was similar (0.620 to 0.635), but recall was higher for rural collisions (0.906) than for urban ones (0.766), and rose from 0.789 at 20 to 30 mph to 0.907 at 60 to 70 mph.

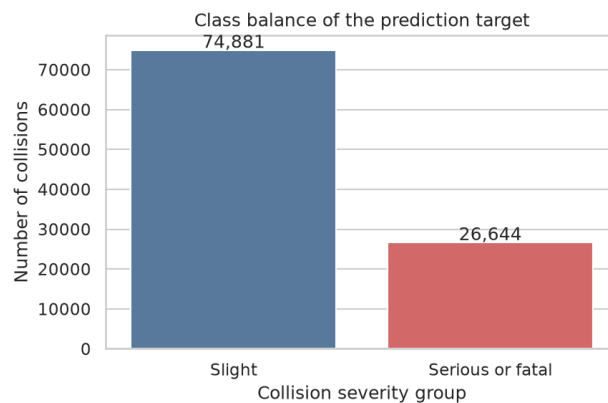


Figure 1. Class balance of the prediction target.

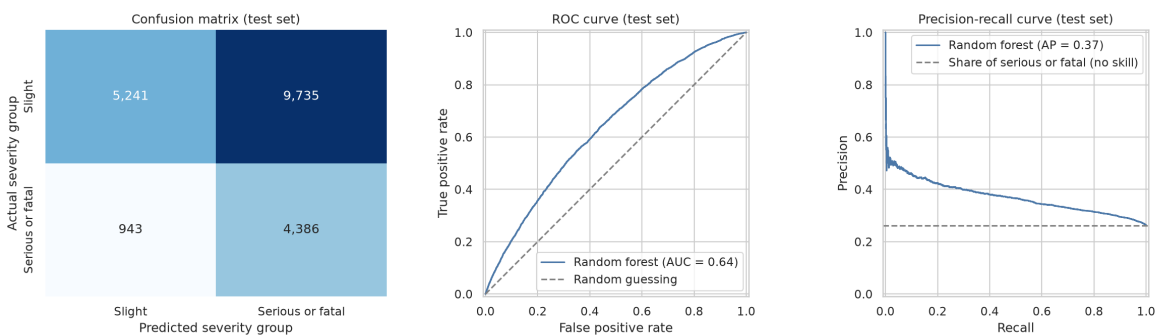


Figure 2. Test-set confusion matrix, ROC curve and precision-recall curve.

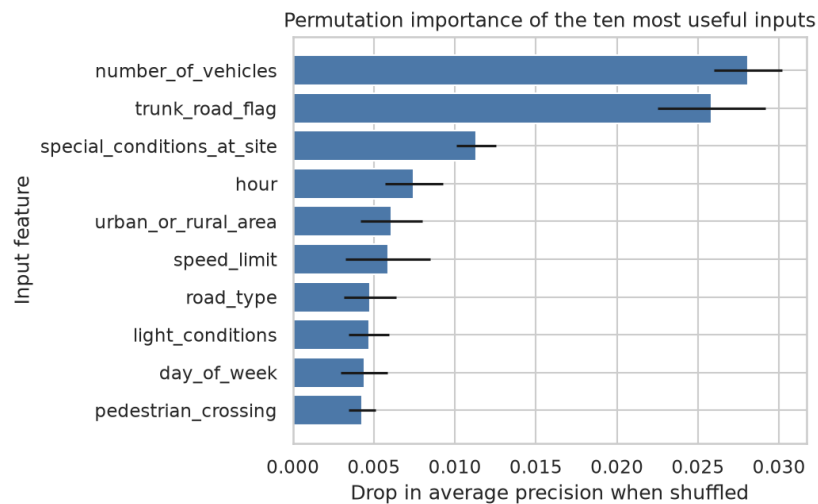


Figure 3. Permutation importance of the ten most useful inputs.

Interpretation for a Non-Technical Audience

I built a computer model that looks at the conditions of a road collision, such as the type of road, the speed limit, the light and the number of vehicles, and estimates whether the collision will turn out to be serious or fatal. About one collision in four is serious or fatal. When the model was tested on collisions it had never seen, it picked out about 82 of every 100 serious or fatal collisions. The price of catching so many is that it raised many false alarms, and only about 31 of every 100 collisions it flagged really were serious or fatal. In other words, the model is better than guessing but far from precise. The conditions at the scene tell us something about how severe a collision is likely to be, but not nearly enough to say what will happen in any one collision, because much depends on things the data do not contain, such as the speed of the vehicles, the type of road user and what protection people had. The model should be seen as a way to explore patterns, and not as a tool for making decisions about individual collisions or places.

Limitations and Potential Bias

Limitations. The inputs carry limited information about severity, which is why ROC AUC is only 0.64 and precision is low. The data cover a single year, and all collisions that were not reported to the police are absent, so the model describes recorded collisions only (Department for Transport, 2026). The label itself is uncertain: the Department notes that the introduction of injury-based reporting of casualty severity by some police forces appears to have changed the reported severity of casualties (Department for Transport, 2026), so the same collision may be recorded differently depending on the force. The model also depends on a threshold that trades missed serious collisions against false alarms, and the right balance is a policy decision and not a statistical one.

Potential bias and responsible use. Bias can enter a machine learning system through the way data are collected and recorded, and models should be examined for uneven performance across groups (Mehrabi et al., 2021). Two sources are relevant here. First, recording practices differ between police forces, and the unknown trunk road flag, which is concentrated in one force code, became one of the model's most useful inputs, so the model partly learns reporting patterns and not only road conditions. Second, with a single threshold the model performs differently for urban and rural collisions and across speed limits, because these groups differ in how common serious collisions are. A model used to allocate road safety funding could therefore direct attention unevenly. To reduce this risk I took two

steps: I excluded location, police force and area identifiers from the inputs, so the model cannot simply learn where serious collisions were previously recorded, and I evaluated performance separately by area type and speed limit. I would further propose repeating the analysis on several years with a time-based split, and comparing group-specific thresholds, before any practical use.

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